Cristian Sbrolli, Ph.D. Candidate

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Education

2017 – 2020 **B.Sc. Computer Science and Engineering, Politecnico di Milano**
Grade: (110/110)

2020 – 2022 **M.Sc. Computer Science and Engineering, Politecnico di Milano**
(110/110 cum Laude)
Thesis title: *Single-view shape reconstruction via Image-Conditioned 3D diffusion.*

2025/05 – 2025/10 **Ph.D. Visiting Student** at The University of Tokyo, Computer Vision and Media Lab. Supervisor: Toshihiko Yamasaki.

2023 – **Ph.D. Computer Science and Engineering, Politecnico di Milano** in Self/Weakly Supervised Multimodal Learning.
Expected Graduation: 2026

Research Publications

Journal Articles

1. N. F. Resmini, E. Lomurno, C. Sbrolli, and M. Matteucci, "Your image generator is your new private dataset," *Image and Vision Computing*, p. 105 727, 2025.

Conference Proceedings

1. G. Savazzi, E. Lomurno, C. Sbrolli, A. Chiatti, and M. Matteucci, "Neuro-symbolic scene graph conditioning for synthetic image dataset generation," 2025, arXiv–2503, to appear in LOD 2025 proceedings.
2. C. Sbrolli and M. Matteucci, "Sceneforge: Enhancing 3d-text alignment with structured scene compositions," 2025, arXiv–2509, to appear in NeurIPS 2025 proceedings.
3. C. Sbrolli, P. Cudrano, and M. Matteucci, "Can shape-infused joint embeddings improve image-conditioned 3d diffusion?" In *2024 International Joint Conference on Neural Networks (IJCNN)*, 2024, pp. 1–10. DOI: 10.1109/IJCNN60899.2024.10649907.
4. C. Sbrolli, M. Matteucci, *et al.*, "No captions, no problem: Captionless 3d-clip alignment with hard negatives via clip knowledge and llms," in *35th British Machine Vision Conference 2024,{BMVC} 2024, Glasgow, UK, November 25-28, 2024*, 2024, pp. 1–10.
5. C. Sbrolli, P. Cudrano, and M. Matteucci, "Cispc: Embedding images and point clouds in a joint concept space by contrastive learning," in *Image Analysis and Processing – ICIAP 2023*, G. L. Foresti, A. Fusiello, and E. Hancock, Eds., Cham: Springer Nature Switzerland, 2023, pp. 468–476, ISBN: 978-3-031-43153-1.
6. L. Lestingi, C. Sbrolli, P. Scarmozzino, G. Romeo, M. M. Bersani, and M. Rossi, "Formal modeling and verification of multi-robot interactive scenarios in service settings," in *Proceedings of the IEEE/ACM 10th International Conference on Formal Methods in Software Engineering*, ser. FormaliSE '22, Pittsburgh, Pennsylvania: Association for Computing Machinery, 2022, pp. 80–90, ISBN: 9781450392877. DOI: 10.1145/3524482.3527653.

Skills

Languages	Italian: First Language English: Cambridge English Level 2 Certificate in ESOL International (FCE First Certificate – 182/190 (Grade A) Council of Europe Level C1 Japanese: Basic Reading and understanding, Politecnico di Milano: Beginner Japanese Course Part 1 (concluded) - Beginner Japanese Course Part 2 (attending)
Coding	Python, Java, C, C++, SQL, LaTeX
NN Frameworks	Pytorch, Pytorch Lightning, Tensorflow, Keras
Databases	MySQL, MongoDB

Additional Professional Activities

Awards and Achievements

2022 **Merit Scholarship**, Politecnico di Milano: Scholarships for particularly deserving off-site students.

2024 **Travel Grant**, IEEE CIS Travel Grant for students presenting a paper being the major contributor.

Extracurricular Activities and Certificates

2022 **Extracurricular Program**. Politecnico di Milano: Honours Programme for Scientific Research in Information Technology.

2023 **6 Months Extracurricular Stage**. Politecnico di Milano, Artificial Intelligence and Robotics Lab (AIRLab): Multimodal Learning for 2D and Point Clouds. Collaboration with the Lab team on 2D Human pose estimation.

M.Sc. Thesis Supervision

2024 Concluded thesis:
3D Scene Understanding in Agricultural Environment.
Semantic and geometric coherence in image to 3d generation using diffusion models.
SAR image despeckling via diffusion models.
Enhancing contrastive learning with synthetic data from text to image models.
Textual Embedding Synthesis to guide Conditioning for Synthetic Dataset Generation in GANs and Stable Diffusion.
Neuro-Symbolic Conditioning for Synthetic Dataset Generation via GANs and Stable Diffusion.

2025 Ongoing thesis:
Federated Multimodal Survival Analysis.

Reviewing

2023 Reviewer for ICIAP 2023, Reviewer for Data & Knowledge Engineering 2024.

2024 Reviewer for BMVC 2024, Reviewer for ICVSE 2024.

Additional Professional Activities (continued)

2025 Reviewer for ICCV 2025, Reviewer for NeurIPS 2025, Reviewer for CVPR 2025, Reviewer for ICLR 2025.